

MOD 306 Course Outline

MOD 306 — CDCGE with EDIS

Course Outline

School	03 Economic Modelling
Level	300 (practitioner / intermediate to advanced)
Hours	45 (indicative band 40–50)
Shape	4 parts · 10 sessions · 7 laboratories · 4 assignments · 1 examination · 1 final project
Model and data	CDCGE - AICTEM Country-Disaggregated CGE (ENVISAGE 10.01 structure) · GTAP 11 (v11b), reference year 2017, values in USD million
Delivery	Instructor-led — online, blended or intensive in person
Prerequisites	MOD 303 and MOD 304. MOD 305 is recommended: the EDIS modules run on a dynamic path, and this course inherits both of that course’s findings. Learners should already read a parameter file before a model and a termination status before a variable.
Assessment	Pass 60 %, with 60 % laboratory competency
Version	1.0 · reviewed 2026-08-27 · 12-month cycle

What this course is

EDIS wires a CGE model to modules that a CGE model does not contain: infrastructure stocks, human development outcomes, education, health, water, sanitation and energy access. This course sets out that architecture — ten modules, nine levers, five macro feedback variables — reads the 37 parameters that govern it, and asks the question the architecture invites: what does a linkage parameter assert, and what would it take to know whether it is right? The extraction carries the register and no output, so the course teaches the wiring and reports honestly that nothing has been run through it.

This is the course with a negative headline. The EDIS extraction contains no coupled result: no module has produced output, no coupled run has been performed and no convergence diagnostic exists, so the course reports the architecture and says plainly that an architecture diagram is not a result. Five of the ten modules would need data this database does not contain, and the course names which and why. Walking the twenty-five declared linkages against the files the manifest points at is a laboratory rather than a claim. All 51 registered findings are in the appendix.

The rule the course is built on

An architecture diagram is not a result. A declared linkage and an implemented linkage look identical in a manifest. The only way to tell them apart is to open what the manifest points at, which takes a minute and is skipped almost every time. The course's own register was empty when it was first read, and that is the finding it is built on rather than the embarrassment it postpones.

Who it is for

Modellers and planners working on an integrated economy-development model, and reviewers asked whether a declared linkage between two models is implemented or merely declared.

Structure

Part	Title	Sessions	Laboratory	Hours
P1	What EDIS is, and how it is put together	01–03	01	9.5
P2	The parameter register and the composites	04–05	02–03	9.0
P3	Human development, and what the database can support	06–08	04–06	13.0
P4	Coupling, convergence and reporting	09–10	07	6.5
	Assignment and assessment time			7.0
	Total	10	7	45.0

The 10 sessions and 7 laboratories, in teaching order, and the subject matter of each:

P1 What EDIS is, and how it is put together — Sessions 01–03 · 9.5 h · Laboratory 01

- **01 What EDIS Is** — What a CGE model cannot represent on its own; The integration problem, stated; Modules, levers and feedback variables; Soft coupling against hard coupling; What the extraction carries: a register, not results; How this course is assessed
- **02 The Module Architecture** — The module list, read one by one; Which are upstream of the CGE and which downstream; BridgeIO: the interface, and why it is separate; What a module needs from the core model; What the core model needs back; Where the architecture could break
- **03 Levers and Feedback** — The nine levers, and what each moves; The five macro feedback variables; Tracing a loop end to end; Open loops and closed loops; Why the feedback set is small; What a lever with no base does — MOD 303, again

- *Lab 01 The Module Register* — What does EDIS declare, and what has it produced?

P2 The parameter register and the composites — Sessions 04–05 · 9.0 h · Laboratory 02–03

- **04 The EDIS Parameter Register** — Every EDIS parameter, by module; What a linkage elasticity is; The roundness diagnostic; What evidence would identify one of these; Which are exactly zero, and what that means; Reporting a linkage parameter honestly
- **05 The Infrastructure Composite** — Roads, rail and water transport; Checking that the weights close; What a composite stock asserts; Physical stocks against service flows; What the composite cannot represent; Where infrastructure data would come from
- *Lab 02 The EDIS Parameter Register* — What are the 37 EDIS parameters, and what does each assert?
- *Lab 03 The Roundness Diagnostic* — Does anything in this parameter file look measured?

P3 Human development, and what the database can support — Sessions 06–08 · 13.0 h · Laboratory 04–06

- **06 The Human Development Block** — The human-development outcome variables; The beta parameters: continuation, entry, passing; Health productivity and the inflection point; The labour feedback coefficient; What a stock-flow outcome model needs; Why this block is the hardest to identify
- **07 Education, Health, Water and Sanitation** — The education linkages and what they drive; The health linkages; Water and sanitation access; Energy access, and its two parameters; The common structure: phi_outcome_driver; Overlap, double counting, and how to check
- **08 What This Database Can Support** — The data each module requires; What the SAM supplies and what it does not; The household boundary — M-05, again; The sector boundary; The spatial boundary; An honest module-by-module verdict
- *Lab 04 Do the Weights Close?* — Does the infrastructure composite sum to one?
- *Lab 05 Trace a Loop* — Follow one lever from the CGE to an outcome and back.
- *Lab 06 What Could Run?* — Which modules could this database support?

P4 Coupling, convergence and reporting — Sessions 09–10 · 6.5 h · Laboratory 07

- **09 Coupling and Convergence** — One-way, iterated and simultaneous coupling; The convergence criterion for a coupled system; What can oscillate, and why; Testing convergence rather than assuming it; How failure would present itself; Inheriting MOD 305's findings
- **10 Reporting an Integrated Model** — The three questions a reader needs answered; Distinguishing architecture from result; The integration note template; How integrated models are oversold; The honest statement for this extraction; What would change it
- *Lab 07 The Integration Note* — What can honestly be said about this integration?

What a graduate can do

1. Set out the EDIS module architecture and say what each module connects.
2. Distinguish a lever from a feedback variable, and trace a loop through both.
3. Read the EDIS parameter register and classify every entry by module.
4. State what a linkage elasticity asserts, and what evidence would identify it.
5. Explain the infrastructure composite and check that its weights close.
6. Set out the human-development block and its outcome variables.
7. Explain how education, health, water and sanitation linkages feed back to the labour force.
8. Identify which modules could run on this database and which could not.
9. Design a two-way coupling and say how convergence would be tested.

10. Diagnose a parameter file for signs that its values were chosen rather than measured.
11. Write an integration note that states what is wired, what is parameterised, and what has been run.

The formal learning outcomes, and the map from each to the assessment that tests it, are in 12_LMS/learning_outcomes.md.

Assessment

Component	Weight	What it is
Knowledge quizzes	—	A bank of 120 items, used for formative practice and drawn on by the theory paper, which carries the marks
Laboratory exercises	35 %	7 laboratories, equally weighted. A laboratory is not complete until its validation checks have been read and its named deliverable produced: an irreproducible result is no result
Applied assignments	20 %	The architecture register; The parameter register and the roundness diagnostic; Trace a loop, find the bridge; The integration note
Theory examination	20 %	Closed-book, drawn from the quiz bank and the review questions
Final project	25 %	A complete review of an integrated modelling framework — this one, or one the learner is responsible for — delivered as a nine-section integration note plus the diagnostic scripts that produced it

Pass 60 % overall and 60 % laboratory competency. Merit 70 %. Distinction 80 %. One resubmission of a single failed component within eight weeks. A laboratory is not complete until every validation check has been read and the provenance of every figure recorded: an irreproducible result is no result.

Data and access

No restricted data. The thirty-seven EDIS parameters carry no provenance in the source, and the course does not supply any: it records that they carry none, which is a different statement from recording their values. Nothing here is a coupled result, and no figure in the course may be read as one.

Materials and set-up

A 110-page Technical Manual, 11 lecture decks of 122 slides, 7 laboratory decks of 90 slides, a Laboratory Manual, Instructor Guide and Learner Workbook, a 120-item quiz bank, 5 assessment files, 15 datasets, 4 code exercises, 3 case studies and the LMS package. The README lists every file, and COURSE_BUILD_REPORT.md records how each was produced.

CDCGE at D:/EDIP/models/CDCGE with the EDIS module tree present, and Julia with `–pkgimages=no`. Laboratory 6 asks what could run rather than running it, and needs the module directory readable but nothing solved.

The 51 findings registered against this school, and the bearing each has on this course, are carried in the Technical Manual's gap appendix and listed in `COURSE_VERSION.json`.

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